An exact spectral backward-error counterexample for LSMR

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11 September 2026

Independently reviewed resolution. The complete proof was checked against the current canonical target IE-17. The recovered provisional identifiers are confirmed. Authorship is recorded at the submitter's request; substantial AI assistance and reconstruction are disclosed in the submission record. This resolves the canonical spectral-norm formulation; no Frobenius-error conclusion is inferred from the cited paper's different default norm convention. The dated independent agent review records the subsequent checks and supersedes original draft remarks about pending review. This is not external human peer review or formal certification.

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

An integer 4×3 least-squares problem has exact LSMR iterates with increasing optimal matrix-only spectral-norm backward error. A spectral completion gives an upper certificate 1979/2000 for the squared first error, and a positive-definite rational matrix gives a strict lower certificate 99/100 for the squared second error. The stated backward-error approximation increases as well. These certificates use exact rational arithmetic, not a floating-point optimizer or substitution of the Frobenius norm.

1 Definitions and exact iterates

For $r = b - Ax$, define

$$\mu(x) = \min\{\|E\|_2 : (A+E)^T((A+E)x - b) = 0\}.$$

Only A is perturbed; the norm is spectral. The approximation uses $\eta = \|r\|_2 \,/\, \|x\|_2$, $K = (A^T, \eta I)^T$, $v = (r^T, 0)^T$, and $\widetilde{\mu}(x) = \left\|KK^\dagger v\right\|_2 / \|x\|_2$. For nonzero x, r, projection algebra gives

$$\widetilde{\mu}(x)^2 = r^T A(\|x\|_2^2 A^T A + \|r\|_2^2 I)^{-1} A^T r.$$

Use no damping, $x_0 = 0$, and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{pmatrix}, \qquad b = (11, 1, 1, 1)^T.$$

The matrix has full column rank. Put $H = A^T A = \mathrm{diag}(1, 36, 25)$ and $g = A^T b = (11, 6, 5)^T$. Exact LSMR minimizes $\|g - Hx\|_2$ over $\mathcal{K}_k(H, g)$. Its first two iterates are

$$x_1 = \frac{1021}{31201} g = \frac{1}{31201}(11231, 6126, 5105)^T,$$

$$x_2 = \frac{16321g - 383Hg}{110438} = \frac{1}{55219}(87659, 7599, 16865)^T.$$

For $V_1 = (g)$ and $V_2 = (g, Hg)$, direct multiplication checks $x_k \in \operatorname{range} V_k$ and $(HV_k)^T(g - Hx_k) = 0$. The columns of HV_k are independent, so these conditions uniquely certify the minimizing iterates. The third is $(11, 1/6, 1/5)^T$, the exact least-squares solution.

The first residuals are

$$r_1 = \frac{1}{31201}(331980, -5555, 5676, 31201)^T, \quad r_2 = \frac{1}{55219}(519750, 9625, -29106, 55219)^T.$$

The normal residuals decrease, as required by LSMR:

$$\left\|A^T r_1\right\|_2^2 = \frac{3593700}{31201} > \frac{5336100}{55219} = \left\|A^T r_2\right\|_2^2.$$

2 General lower and upper certificates

For nonzero x, write $s = \|x\|_2^2$, $z = Ax$, $r = b - z$, and

$$C = AA^T, \qquad D = \frac{\|r\|_2^2 I - rr^T + zz^T}{s}.$$

Lemma 1. *If $m > n$ and $0 \le t \le 1$, then $\mu(x)^2 \ge \lambda_{\min}(tC + (1-t)D)$.*

Proof. For a feasible E whose new residual is nonzero, let u be its unit direction. Then $(A+E)^T u = 0$, so $\|E\|_2^2 \ge u^T C u$. The residual is $uu^T b$, because it is orthogonal to $(A + E)x$. Therefore $Ex = r - uu^T b$, and $\|E\|_2^2 \ge \left\|r - uu^T b\right\|_2^2 / s = u^T D u$. A convex combination proves the claim.

For zero new residual, $Ex = r$, so $\|E\|_2^2 \ge \|r\|_2^2 / s$. Choose unit $u \in \ker A^T$, possible since $m > n$. Then $u^T z = 0$ and

$$u^T[tC + (1-t)D]u = (1-t)(\|r\|_2^2 - (u^T r)^2)/s \le \|r\|_2^2 / s.$$

This proves the same bound in that case. □

Lemma 2. *If a unit vector u has $u^T C u < \kappa$ and $u^T D u < \kappa$, with $\kappa > 0$, then a feasible perturbation has $\|E\|_2 \le \sqrt{\kappa}$.*

Proof. Let $e = x/\sqrt{s}$ and prescribe

$$E^T u = -A^T u =: de + a_\perp, \qquad Ee = (r - uu^T b)/\sqrt{s} =: du + c_\perp,$$

where $a_\perp \perp e$, $c_\perp \perp u$, and the shared scalar is $d = -u^T Ax/\sqrt{s}$. The two squared lengths are $u^T C u, u^T D u$. A completion is

$$E = due^T + ua_\perp^T + c_\perp e^T - \frac{d}{\kappa - d^2} c_\perp a_\perp^T.$$

To bound its norm, put $d_0 = d/\sqrt{\kappa}$, $r_0 = \sqrt{1 - d_0^2}$. In decompositions along u, e, $E/\sqrt{\kappa}$ factors as

$$\begin{pmatrix} 1 & 0 \\ 0 & c_\perp/(\sqrt{\kappa} r_0) \end{pmatrix} \begin{pmatrix} d_0 & r_0 \\ r_0 & -d_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & a_\perp^T/(\sqrt{\kappa} r_0) \end{pmatrix}.$$

The outside maps have norm at most one by the prescribed length bounds, and the middle map is orthogonal. The prescriptions give $(A+E)^T u = 0$ and $(A+E)x = (I - uu^T)b$, proving feasibility. □

3 Strict increase

For x_1, choose $w = (250, -1, 1, 27)^T$, $u = w/\sqrt{63231}$, $\kappa = 1979/2000$. Exact evaluation gives

$$u^T C u = \frac{62561}{63231} < \kappa, \qquad u^T D u = \frac{1642993919237}{1713779258646} < \kappa.$$

The upper lemma applies. For x_2, set $t = 5/6$ and $c = 99/100$. Then $tC + (1-t)D - cI = K/2407881992100$, where

$$K = \begin{pmatrix} 206417059721 & -50293465200 & 1125984433750 & -1435003762500 \\ -50293465200 & 83658415217471 & 206242965000 & -26574143750 \\ 1125984433750 & 206242965000 & 61800032332121 & 80360210700 \\ -1435003762500 & -26574143750 & 80360210700 & 11170189945871 \end{pmatrix}.$$

The four leading principal determinants are

$$\begin{aligned}
&206417059721,\\
&17265994657467102998545591,\\
&9609413247404803317937438451780862910 11,\\
&65442104145157248520714038046591467785805073713081.
\end{aligned}$$

All are positive. Sylvester's criterion and the lower lemma give

$$\boxed{\mu(x_1)^2 \le \frac{1979}{2000} < \frac{99}{100} < \mu(x_2)^2.}$$

4 Approximation and rational witness

The squared approximations are exactly

$$\begin{aligned}
\widetilde{\mu}(x_1)^2 &= \frac{6969410785257343950389 2031925}{69323394392991282508138323472},\\
\widetilde{\mu}(x_2)^2 &= \frac{5430772101137459612205263871781350}{5387955615790281743396033884265233}.
\end{aligned}$$

Cross multiplication yields

$$\boxed{\widetilde{\mu}(x_1)^2 < 1.006 < 1.007 < \widetilde{\mu}(x_2)^2,}$$

where the decimals are exact rational cutoffs.

The upper completion also has a fully rational form. Put $\omega = w^T w$, $h = w^T z$, $c_0 = r - w(w^T r)/\omega$, $a_0 = -A^T w + (h/s)x$. Then

$$E = -ww^T A/\omega + c_0 x^T/s + \frac{h\, c_0 a_0^T}{\omega s \kappa - h^2}.$$

The verifier directly checks the perturbed normal equations and positivity of all leading principal minors of $\kappa I - E^T E$. The complete rational matrix is saved in its results file.

A dense variant multiplies A, b by the Hadamard matrix with rows $(1,1,1,1)$, $(1,-1,1,-1)$, $(1,1,-1,-1)$, $(1,-1,-1,1)$, producing

$$A' = \begin{pmatrix} 1 & 6 & 5 \\ 1 & -6 & 5 \\ 1 & 6 & -5 \\ 1 & -6 & -5 \end{pmatrix}, \qquad b' = (14, 10, 10, 10)^T.$$

This multiplier is twice orthogonal. Iterates are unchanged and both error quantities double, preserving the strict increases without a zero row.

5 Sources and scope

Recovered references: OpenProblemsInNLA, provisionally IE-17, *Monotonic optimal backward error along LSMR*; D. C.-L. Fong and M. A. Saunders, *LSMR: An Iterative Algorithm for Sparse Least-Squares Problems*, SIAM J. Sci. Comput. 33(5), 2011, 2950–2971, DOI 10.1137/10079687X.[1] The target addressed here uses the explicit spectral-norm definition in Section 1. The current repository statement was not independently retrieved during reconstruction. A different norm or a perturbation model allowing changes to b is outside this certificate's stated claim.

[1] `https://arxiv.org/abs/1006.0758`. Reference metadata was retained from earlier notes; source conventions and priority require verification.